DOKUMENTASI TUGAS PROYEK DATA ENGINEERING

TMDB Movie Pipeline

End-to-End Data Pipeline: Bronze → Silver → Gold → Dashboard

Tools & Teknologi
PostgreSQL · dbt · PySpark · Apache NiFi · Apache Kafka · Debezium
ClickHouse · MinIO · Streamlit · Docker · Python

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Link projek di github : https://github.com/nabilahlw/tmdb-pipeline-project.git

Gambaran Umum Proyek

Proyek ini membangun pipeline data end-to-end untuk dataset film TMDB (The Movie Database) menggunakan arsitektur Medallion (Bronze → Silver → Gold). Data bersumber dari dua tempat: dataset Kaggle (4.803 film) dan TMDb API live.

Sumber Data:

TMDb API Live — https://www.themoviedb.org/subscribe/developer

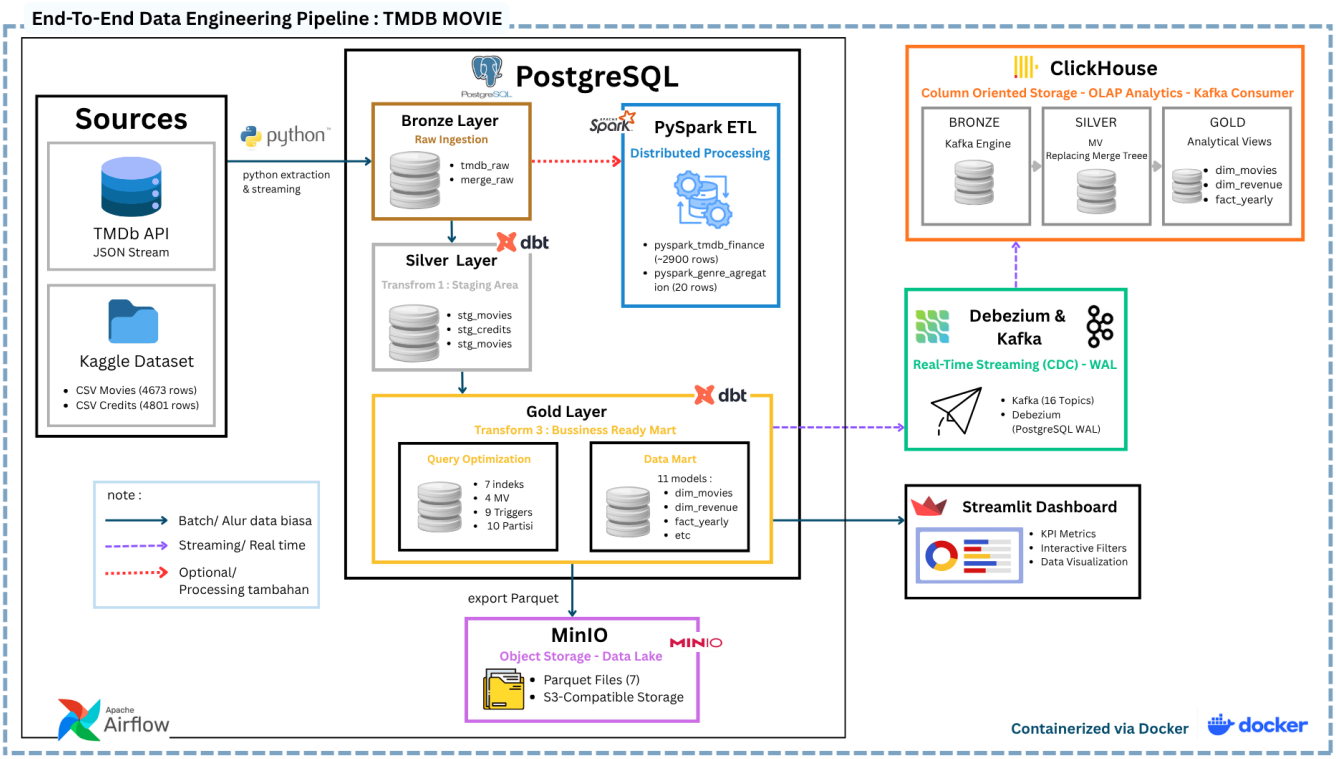
Kaggle Dataset — https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata

Ringkasan Pipeline

Step	Komponen	Output
Step 1	Bronze Layer — Ingestion	tmdb_raw (80 rows), merge_raw (4.803 rows)
Step 2	Silver/Gold — dbt Transformation	11 mart model di schema silver_mart
Step 3	Query Optimization	4 MV, 10 partisi, 7 index, 9 triggers
Step 4	PySpark ETL	pyspark_tmdb_finance, pyspark_genre_aggregation
Step 5	Apache NiFi Visual Pipeline	nifi_tmdb_movie
Step 6	Kafka + Debezium CDC	16 topics, streaming real-time
Step 7	ClickHouse OLAP	9 objek, 9.604 rows dim_movies
Step 8	MinIO Object Storage	7 file Parquet, 2.7 MiB
Step 9	Streamlit Dashboard	Dashboard interaktif dengan filter
Step 10	Apache Airflow / DAG Orchestration	Otomasi 4 step (API, CSV ke PostgreSQL dan MinIO)

Arsitektur Pipeline

Pipeline menggunakan pola EtLT (Extract → Load → Transform → Load → Transform) modern, di mana data dikumpulkan dulu dalam bentuk raw lalu ditransformasi secara bertahap mengikuti Medallion Architecture.



Sebelum menjalankan proyek ini semua container docker yang terlibat harus terbuka.

Name	Container ID	Image	Port(s)	CPU (	Actions
tmdb-pipeline-project	-	-	-	N	
tmdb_airflow_scheduler	33c49eaa546f	apache/airflow:2.8.1		N	
tmdb_airflow_web	457fb22a156e	apache/airflow:2.8.1	8082:8080	N	
tmdb_airflow_init	de3683fc0b9a	apache/airflow:2.8.1		N	
tmdb_debezium	a8c8d14f0c42	debezium/connect:2.4	8083:8083	N	
tmdb_pgadmin	c0e92ce2849b	dpage/pgadmin4	5050:80	N	
tmdb_kafka	17fcf1c8b125	confluentinc/cp-kafka:7.4.0	29092:29092	N	
tmdb_kafka_man	8a0a6fc83902	hlebalbau/kafka-manager:latest	9000:9000	N	
tmdb_postgres	22827ae4eea3	postgres:13	5432:5432	N	
tmdb_clickhouse	4ce4f4eac5e7	clickhouse/clickhouse-server:23.8	18123:8123 Show all ports (2)	N	
tmdb_zookeeper	7dda08543494	confluentinc/cp-zookeeper:7.4.0		N	
tmdb_minio	4a23ad4f533a	minio/minio:latest	9002:9000 Show all ports (2)	N	

Step 1 — Bronze Layer: Environment Setup & Data Ingestion

Bronze layer adalah lapisan pertama dalam Medallion Architecture yang menyimpan data raw tanpa transformasi apapun. Sumber data berasal dari dua tempat yang kemudian digabungkan.

1.1 Infrastruktur Docker (Container)

Container	Image	Port	Fungsi
tmdb_postgres	postgres:13	5432:5432	Database utama PostgreSQL
tmdb_pgadmin	dpage/pgadmin4	5050:80	GUI monitoring database

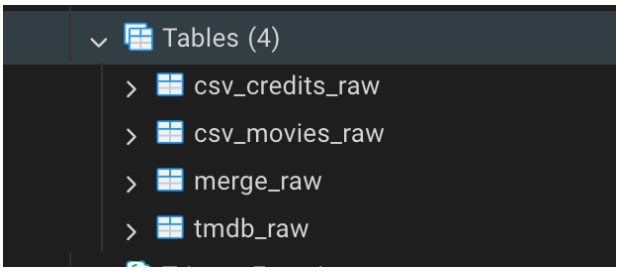
Konfigurasi koneksi pgAdmin: Host = postgres (nama container, BUKAN localhost) | Port = 5432 | Username = admin | Password = adminadmin | Database = tmdb_db

1.2 Sumber Data

Sumber	Format	Jumlah Data	Keterangan
TMDb API live (tmdb_raw)	JSON	80 rows	Endpoint /movie/popular, authentication via API key
Kaggle CSV — tmdb_5000_movies	CSV	4.673 rows	Data historis film dengan kolom budget, revenue, genres
Kaggle CSV — tmdb_5000_credits	CSV	4.801 rows	Data cast dan crew per film

1.3 Struktur Bronze di PgAdmin

Masuk ke http://localhost:5050/browser/
Nama projek : tmdb_postgres, lalu tmdb_db → Schemas → public → Tables → merge_raw, tmdb_raw, csv_movies_raw



1.4 Struktur Tabel tmdb_raw (15 Kolom) dari API Tmdb

Kolom	Tipe Data	Penjelasan

id	bigint	ID unik film dari TMDb
title	text	Judul film dalam bahasa Inggris
original_title	text	Judul asli film
original_language	text	Kode bahasa (en, fr, ko, dll.)
overview	text	Deskripsi / sinopsis film
genre_ids	text	List ID genre (masih bentuk string JSON)
release_date	text	Tanggal rilis (format: YYYY-MM-DD)
popularity	double precision	Skor popularitas dari TMDb
vote_average	double precision	Rating rata-rata (skala 0-10)
vote_count	bigint	Jumlah vote yang masuk
adult	boolean	Konten dewasa (True/False)
video	boolean	Apakah berupa video (biasanya False)
backdrop_path	text	Path gambar backdrop
poster_path	text	Path poster film
category	text	Kategori (popular/top_rated/dll.)

1.4 Struktur Tabel merge_raw (22 Kolom, 4.803 rows)

Dua file CSV digabungkan menjadi tabel merge_raw dengan shape (4803, 22). Kolom-kolom yang tersedia:

budget, genres, homepage, id, keywords, original_language, original_title, overview, popularity, production_companies, production_countries, release_date, revenue, runtime, spoken_languages, status, tagline, title, vote_average, vote_count, cast, crew

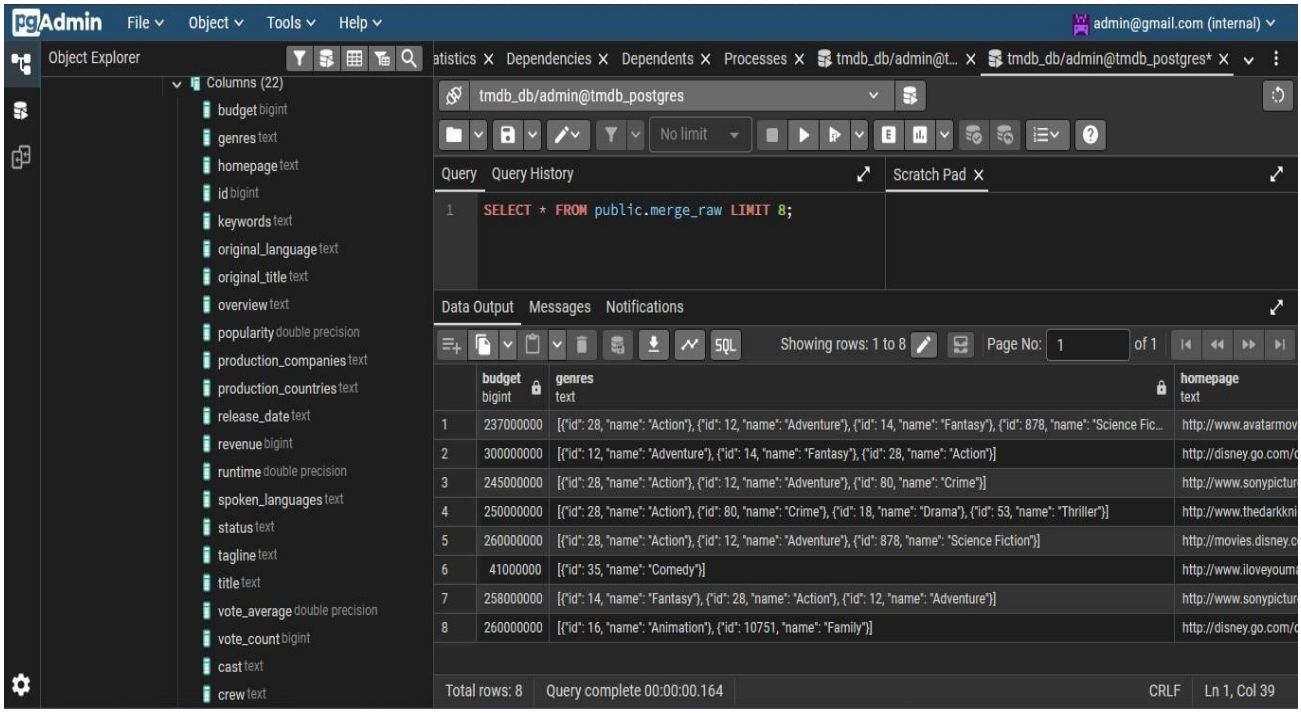
1.5 Kolom JSON yang Perlu Di-parse

NOTE: Kolom-kolom berikut masih dalam format STRING JSON dan akan di-parse di step selanjutnya.

Kolom JSON	Contoh Isi	Di-parse di Step
genres	[{"id":28,"name":"Action"},...]	Step 2 (dbt) & Step 4 (PySpark)
cast	[{"name":"Sam Worthington",...}]	Step 2 (dbt) & Step 4 (PySpark)
crew	[{"job":"Director","name":"..."}]	Step 2 (dbt) & Step 4 (PySpark)
keywords	[{"id":1,"name":"avatar"},...]	Step 2 (dbt)
production_companies	[{"name":"Warner Bros"},...]	Step 2 (dbt) & Step 5 (NiFi)
production_countries	[{"iso":"US","name":"..."},...]	Step 2 (dbt)

spoken_languages	[{"iso":"en","name":"English"},...]	Step 2 (dbt)

Validasi tabel merge_raw di Postgres



Step 2 — Silver & Gold Layer: dbt Transformation

dbt (data build tool) digunakan untuk mentransformasi data dari Bronze layer menjadi tabel-tabel analitik yang bersih dan terstruktur. Arsitektur menggunakan 2 layer: Staging (pembersihan) dan Mart (analitik).

Letak di PgAdmin = tmdb_db → Schemas → **silver_mart_silver_mart** → 11 Tables (dim_movies, dim_revenue, dim_genres, dll)
→ **Views** (stg_movies, stg_credits, stg_tmdbapi)

2.1 Environment dbt

Komponen	Nilai
dbt version	1.11.7 (latest)
Python version	3.11.9
PostgreSQL adapter	dbt-postgres 1.10.0 — OK
Target schema	silver_mart
Jumlah model	3 staging + 11 mart = 14 model total

2.2 Staging Layer (3 Model)

Model	Sumber	Transformasi Utama
stg_movies.sql	merge_raw	Handle 2 format tanggal (DD/MM/YYYY & YYYY-MM-DD), parse semua kolom JSON, normalisasi popularity & vote_average
stg_credits.sql	merge_raw	Ekstrak lead actor, top 3 cast, director, screenplay writer, producer, cinematographer, music composer dari JSON
stg_tmdbapi.sql	tmdb_raw	Fix format tanggal, handle genre_ids sebagai PostgreSQL native array

2.3 Mart Layer (11 Model) — Schema: silver_mart

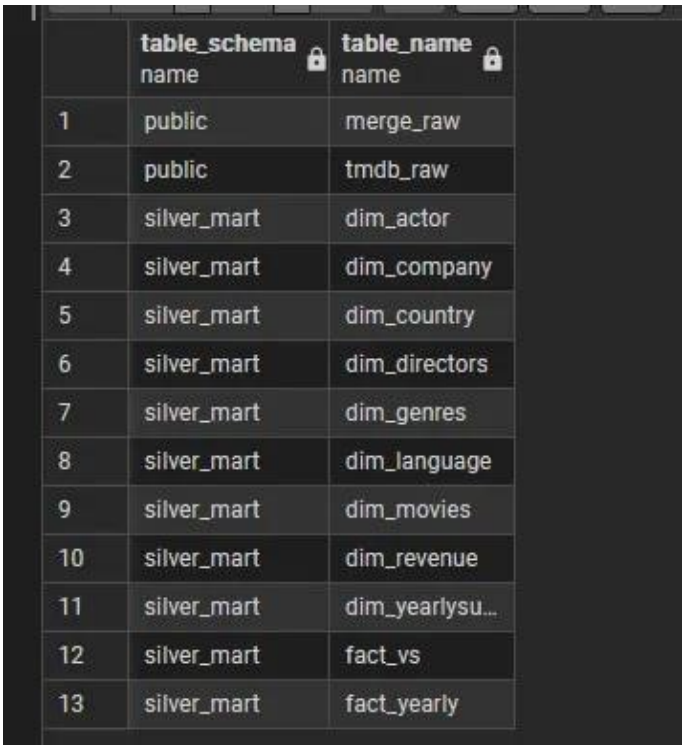
Model	Deskripsi	Kolom Kunci
dim_movies	Tabel master film lengkap, gabungan 3 sumber	movie_id, title, primary_genre, director, lead_actor, release_year
dim_genres	Ringkasan performa per genre	genre, total_films, total_revenue, avg_rating, top_grossing_film
dim_directors	Profil karir sutradara	director, total_films, director_roi_pct, filmography
dim_actor	Profil karir aktor (lead + supporting)	actor_name, lead_rate_pct, frequent_director, best_performing_film
dim_country	Ringkasan per negara produksi	primary_country, total_films, most_common_genre, most_active_director
dim_language	Analisis per bahasa film	original_language, total_films, dominant_country, top_revenue_genre
dim_company	Profil rumah produksi	main_production_company, blockbuster_rate_pct, most_active_director
dim_revenue	Fakta keuangan per film	performance_label, roi_pct, popularity_tier, profit
fact_vs	Rekonsiliasi 2 sumber data (CSV vs API)	in_api, in_kaggle, popularity_diff, api_vote_average
fact_yearly	Tren industri per tahun per genre	release_year, primary_genre, blockbuster_count, loss_count
dim_yearlysum	Ringkasan industri per tahun (1 baris/tahun)	most_active_director, top_grossing_film, most_common_genre

2.4 Kategori Performa Film (dari mart layer dim_revenue)

Label	Kriteria	Keterangan
Mega Blockbuster	revenue ≥ budget × 3	Film sukses besar, balik modal 3x lipat
Blockbuster	revenue ≥ budget × 2	Film sukses, balik modal 2x lipat
Profitable	revenue > budget	Untung tapi tidak signifikan

Break Even	revenue = budget	Tidak untung tidak rugi
Loss	revenue < budget	Film merugi

Memastikan 11 tabel mart sudah ada di silver_mart plus 2 tabel bronze di public di PgAdmin https://github.com/nabilahlw/tmdb-pipeline-project.git



	table_schema name	table_name name
1	public	merge_raw
2	public	tmdb_raw
3	silver_mart	dim_actor
4	silver_mart	dim_company
5	silver_mart	dim_country
6	silver_mart	dim_directors
7	silver_mart	dim_genres
8	silver_mart	dim_language
9	silver_mart	dim_movies
10	silver_mart	dim_revenue
11	silver_mart	dim_yearlysu...
12	silver_mart	fact_vs
13	silver_mart	fact_yearly

Step 3 Query Optimization: EXPLAIN, Index, Partition, MV

Step 3 membuktikan kemampuan optimasi performa database menggunakan 4 teknik utama: EXPLAIN untuk profiling, Index untuk mempercepat lookup, Partition untuk membagi data besar, dan Materialized View sebagai cache agregasi.

Letak Optimization di PdgAdmin : tmdb_db → Schemas → silver_mart_silver_mart → Tables → table dim_movies → klik kanan → Properties → Indexes (lihat 7 index di sana)

3.1 Hasil Validasi Akhir

Objek	Jumlah	Detail
Indexes	7	idx_dim_movies_director, idx_dim_movies_genre, idx_dim_movies_year, idx_dim_movies_genre_rating, + 3 lainnya
Materialized Views	4	mv_genre_summary, mv_revenue_summary, mv_director_performance, mv_country_performance

Triggers	9	3 fungsi auto-refresh × 3 event (INSERT/UPDATE/DELETE) = 9 trigger
Partitions	10	fact_yearly_1920s hingga fact_yearly_2010s (per dekade)

3.2 EXPLAIN — Analisis Performa Query

EXPLAIN dijalankan pada dim_movies dan fact_yearly sebagai baseline sebelum optimasi. Output menunjukkan Seq Scan (membaca seluruh tabel) yang kemudian berubah menjadi Index Scan setelah index dibuat — cost turun drastis.

3.3 Partisi fact_yearly — Per Dekade

Tabel fact_yearly dibagi menjadi 10 sub-tabel berdasarkan release_year menggunakan PARTITION BY RANGE. Ketika query filter WHERE release_year = 2010, PostgreSQL hanya membaca partisi fact_yearly_2010s — bukan seluruh tabel.

	QUERY PLAN text
1	Sort (cost=13.24..13.31 rows=27 width=36)
2	Sort Key: release_year
3	-> HashAggregate (cost=12.26..12.60 rows=27 width=36)
4	Group Key: release_year
5	-> Seq Scan on fact_yearly (cost=0.00..10.45 rows=363 w...
6	Filter: ((release_year >= 1990) AND (release_year <= 20...

Lalu membuat indeks

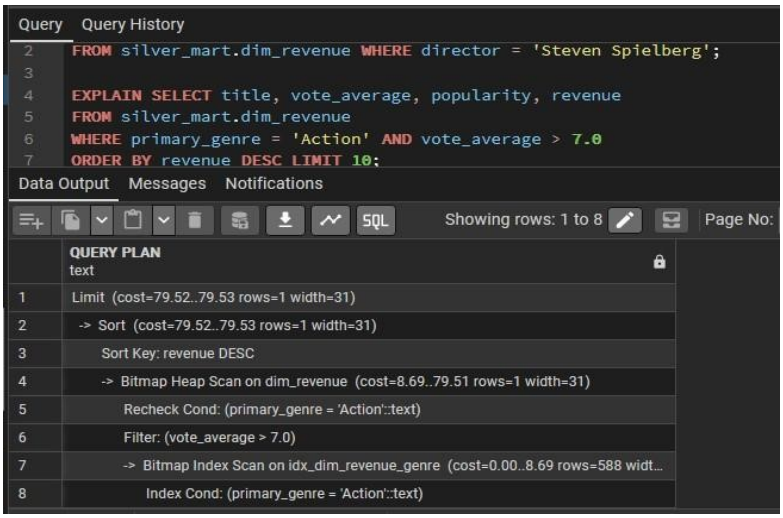
Query | Query History

```
1  CREATE INDEX idx_dim_movies_director    ON silver_mart.dim_movies (director);
2  CREATE INDEX idx_dim_movies_genre       ON silver_mart.dim_movies (primary_genre);
3  CREATE INDEX idx_dim_movies_year        ON silver_mart.dim_movies (release_year);
4  CREATE INDEX idx_dim_movies_genre_rating ON silver_mart.dim_movies (primary_genre, vote_average DESC);
5  CREATE INDEX idx_dim_revenue_genre      ON silver_mart.dim_revenue (primary_genre);
6  CREATE INDEX idx_dim_revenue_year       ON silver_mart.dim_revenue (release_year);
7  CREATE INDEX idx_dim_revenue_perf       ON silver_mart.dim_revenue (performance_label);
```

Data Output | Messages | Notifications

```
CREATE INDEX

Query returned successfully in 381 msec.
```

Membuktikan indeks bekerja, COST turun

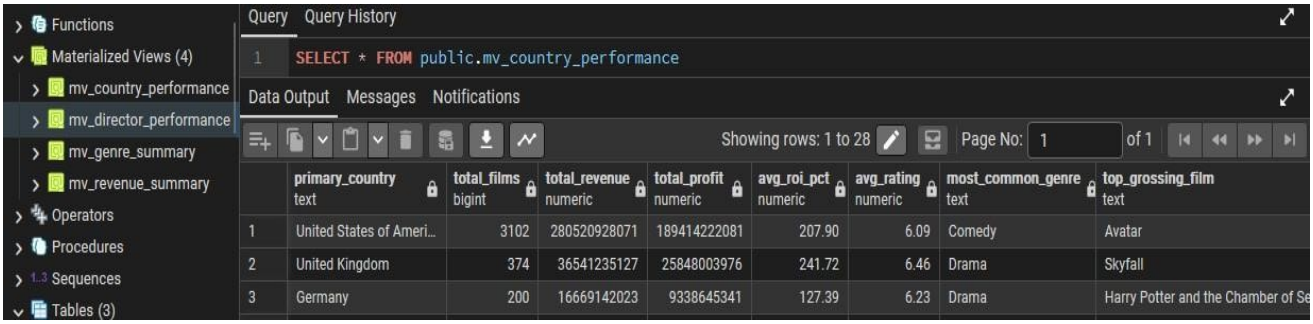


3.4 Materialized View + Auto-Refresh Trigger

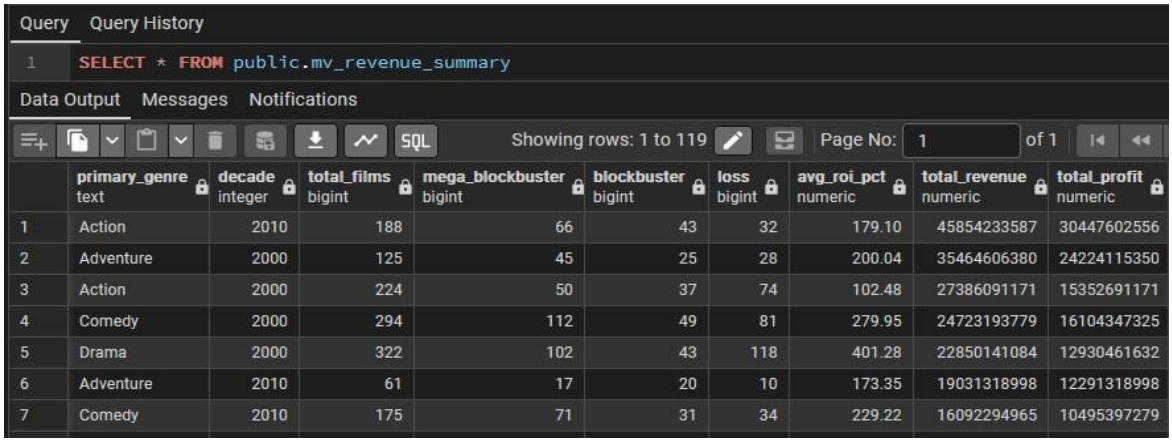
4 Materialized View dibuat sebagai cache agregasi untuk dashboard. Setiap perubahan data di tabel sumber secara otomatis men-trigger REFRESH MV tanpa perlu manual.

Materialized View	Sumber	Insight yang Disediakan
mv_genre_summary	dim_genres	Genre paling cuan, paling diminati, tren tahunan
mv_revenue_summary	dim_revenue	Genre booming per dekade, tren industri film
mv_director_performance	dim_directors	Sutradara paling profitable, durasi karir
mv_country_performance	dim_country	Negara dominan per genre, genre favorit per negara

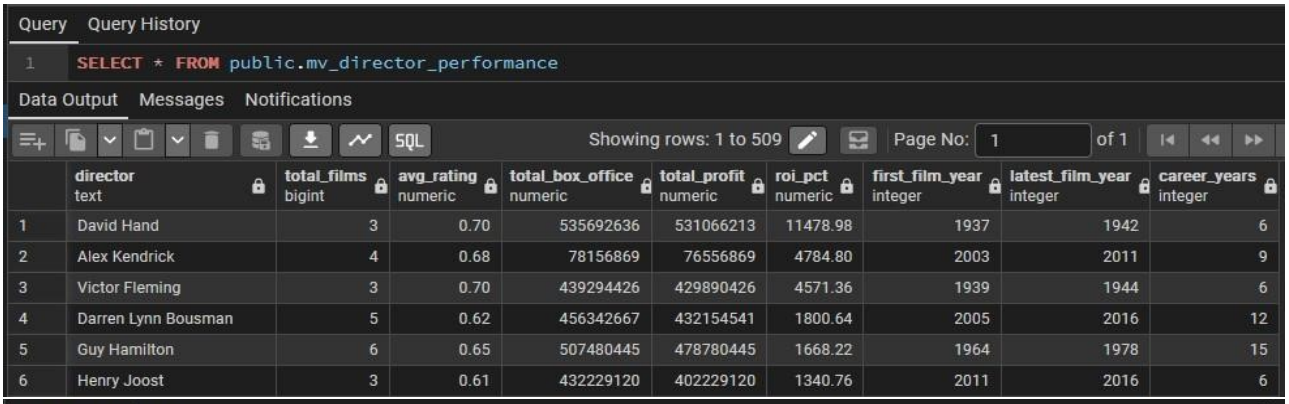
Hasil Validasi MV



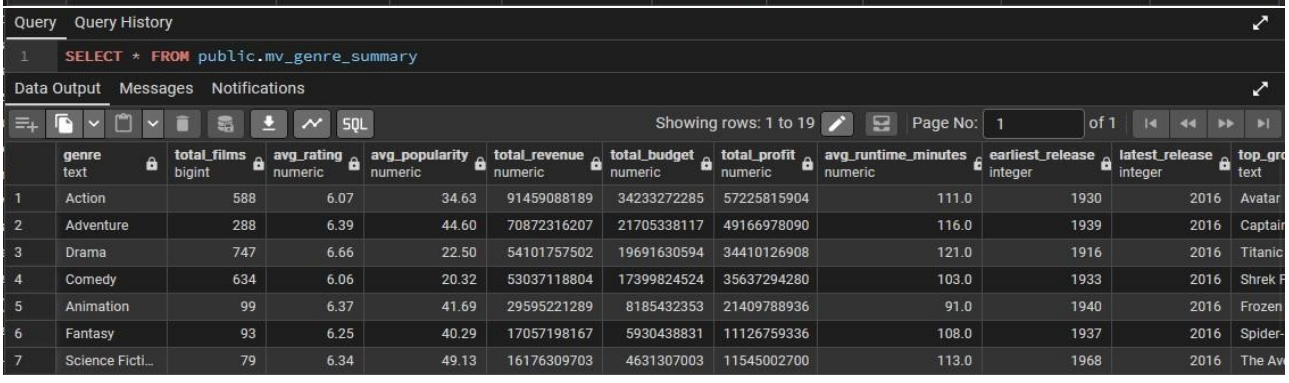
	primary_country text	total_films bigint	total_revenue numeric	total_profit numeric	avg_roi_pct numeric	avg_rating numeric	most_common_genre text	top_grossing_film text
1	United States of Ameri...	3102	280520928071	189414222081	207.90	6.09	Comedy	Avatar
2	United Kingdom	374	36541235127	25848003976	241.72	6.46	Drama	Skyfall
3	Germany	200	16669142023	9338645341	127.39	6.23	Drama	Harry Potter and the Chamber of Se



	primary_genre text	decade integer	total_films bigint	mega_blockbuster bigint	blockbuster bigint	loss bigint	avg_roi_pct numeric	total_revenue numeric	total_profit numeric
1	Action	2010	188	66	43	32	179.10	45854233587	30447602556
2	Adventure	2000	125	45	25	28	200.04	35464606380	24224115350
3	Action	2000	224	50	37	74	102.48	27386091171	15352691171
4	Comedy	2000	294	112	49	81	279.95	24723193779	16104347325
5	Drama	2000	322	102	43	118	401.28	22850141084	12930461632
6	Adventure	2010	61	17	20	10	173.35	19031318998	12291318998
7	Comedy	2010	175	71	31	34	229.22	16092294965	10495397279



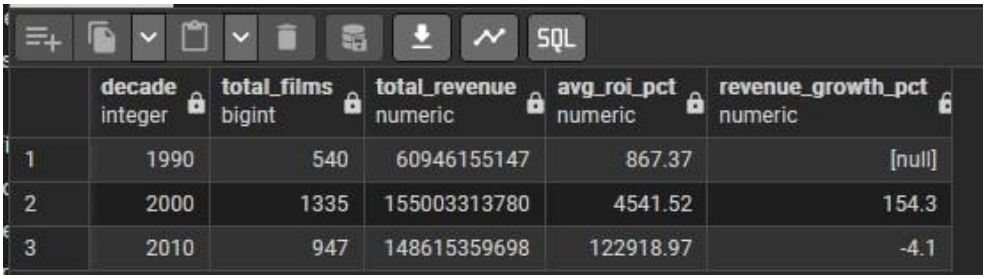
	director text	total_films bigint	avg_rating numeric	total_box_office numeric	total_profit numeric	roi_pct numeric	first_film_year integer	latest_film_year integer	career_years integer
1	David Hand	3	0.70	535692636	531066213	11478.98	1937	1942	6
2	Alex Kendrick	4	0.68	78156869	76556869	4784.80	2003	2011	9
3	Victor Fleming	3	0.70	439294426	429890426	4571.36	1939	1944	6
4	Darren Lynn Bousman	5	0.62	456342667	432154541	1800.64	2005	2016	12
5	Guy Hamilton	6	0.65	507480445	478780445	1668.22	1964	1978	15
6	Henry Joost	3	0.61	432229120	402229120	1340.76	2011	2016	6



	genre text	total_films bigint	avg_rating numeric	avg_popularity numeric	total_revenue numeric	total_budget numeric	total_profit numeric	avg_runtime_minutes numeric	earliest_release integer	latest_release integer	top_gr... text
1	Action	588	6.07	34.63	91459088189	34233272285	57225815904	111.0	1930	2016	Avatar
2	Adventure	288	6.39	44.60	70872316207	21705338117	49166978090	116.0	1939	2016	Captain
3	Drama	747	6.66	22.50	54101757502	19691630594	34410126908	121.0	1916	2016	Titanic
4	Comedy	634	6.06	20.32	53037118804	17399824524	35637294280	103.0	1933	2016	Shrek F
5	Animation	99	6.37	41.69	29595221289	8185432353	21409788936	91.0	1940	2016	Frozen
6	Fantasy	93	6.25	40.29	17057198167	5930438831	11126759336	108.0	1937	2016	Spider-
7	Science Ficti...	79	6.34	49.13	16176309703	4631307003	11545002700	113.0	1968	2016	The Av

3.5 CTE Analitik

CTE (Common Table Expressions) digunakan untuk membuat query lebih mudah dibaca, dengan 3 insight analitik siap pakai: top film per genre, sutradara paling konsisten, dan tren industri per dekade.



	decade integer	total_films bigint	total_revenue numeric	avg_roi_pct numeric	revenue_growth_pct numeric
1	1990	540	60946155147	867.37	[null]
2	2000	1335	155003313780	4541.52	154.3
3	2010	947	148615359698	122918.97	-4.1

Step 4 — PySpark ETL: Distributed Data Processing

PySpark digunakan untuk memproses data merge_raw dengan pendekatan distributed in-memory processing — berbeda dari pandas yang hanya bisa berjalan di satu mesin. Ini membuktikan kemampuan menangani data di skala yang pandas tidak mampu.

4.1 Perbandingan: pandas vs PySpark

Aspek	pandas	PySpark
Eksekusi	Eager — langsung diproses	Lazy — baru eksekusi saat .show()/.write()
Batas data	Maks RAM laptop (~8-32 GB)	Petabyte (skala cluster)
SQL support	Tidak langsung	spark.sql() — full SQL
Cocok untuk	Eksplorasi, prototyping	Production pipeline, data besar

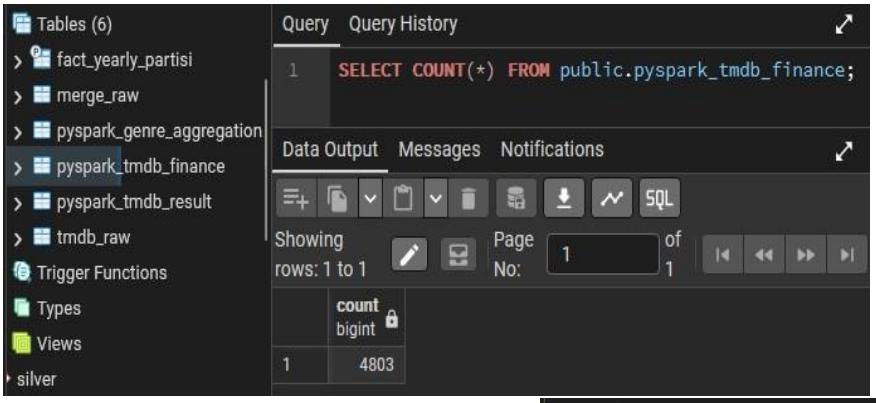
4.2 Alur ETL PySpark

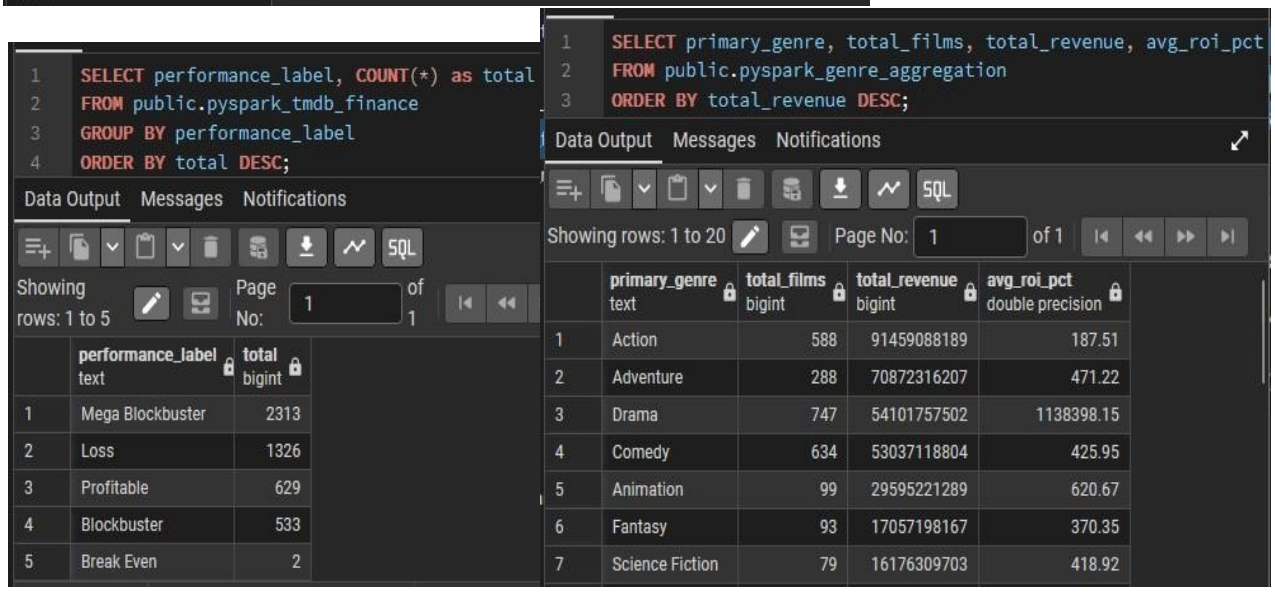
```
EXTRACT  → merge_raw dari PostgreSQL via JDBC (4.803 baris, 22 kolom)

TRANSFORM 1 — Ekstraksi kolom JSON:
  genres → primary_genre
  production_companies → primary_company
  spoken_languages → primary_language

TRANSFORM 2 — Kalkulasi kolom baru:
  profit = revenue - budget
  roi_pct = (revenue - budget) / budget × 100
  performance_label: Mega Blockbuster / Blockbuster / Profitable / Break Even / Loss
  popularity_tier: Viral (>100) / Popular (>50) / Known (>20) / Niche
```

```
TRANSFORM 3 — Agregasi per genre (groupBy primary_genre):
  total_films, avg_rating, total_revenue, total_profit, avg_roi_pct
  mega_blockbuster_count, loss_count

LOAD 1 → public.pyspark_tmdb_finance    (data per film, lengkap)
LOAD 2 → public.pyspark_genre_aggregation (agregasi per genre, 20 rows)
```

4.3 Hasil Validasi

Tabel	Jumlah Rows	Keterangan
pyspark_tmdb_finance	~2.900	Film dengan budget > 0 DAN revenue > 0
pyspark_genre_aggregation	20	1 baris per genre (Action, Drama, Comedy, dll.)





Step 6 — Apache Kafka + Debezium CDC: Real-Time Streaming

Step 6 membangun sistem Change Data Capture (CDC) yang menangkap setiap perubahan data di PostgreSQL secara real-time dan mengalirkannya ke Kafka. Ini adalah pola arsitektur yang digunakan di skala besar seperti Uber (130 juta event/detik).

Membuka Cluster di Kafka : http://localhost:9000

6.1 Komponen Stack Kafka

Container	Image	Port	Fungsi
tmdb_zookeeper	confluentinc/cp-zookeeper:7.4.0	2181	Koordinasi Kafka cluster
tmdb_kafka	confluentinc/cp-kafka:7.4.0	29092:29092	Message broker, penyimpan event stream
tmdb_debezium	debezium/connect:2.4	8083:8083	CDC connector PostgreSQL → Kafka
tmdb_kafka_man	hlebalbau/kafkamanager	9000:9000	Web UI monitoring Kafka

6.2 Konfigurasi WAL PostgreSQL

Untuk Debezium bisa menangkap perubahan, PostgreSQL harus diset ke mode WAL logical:

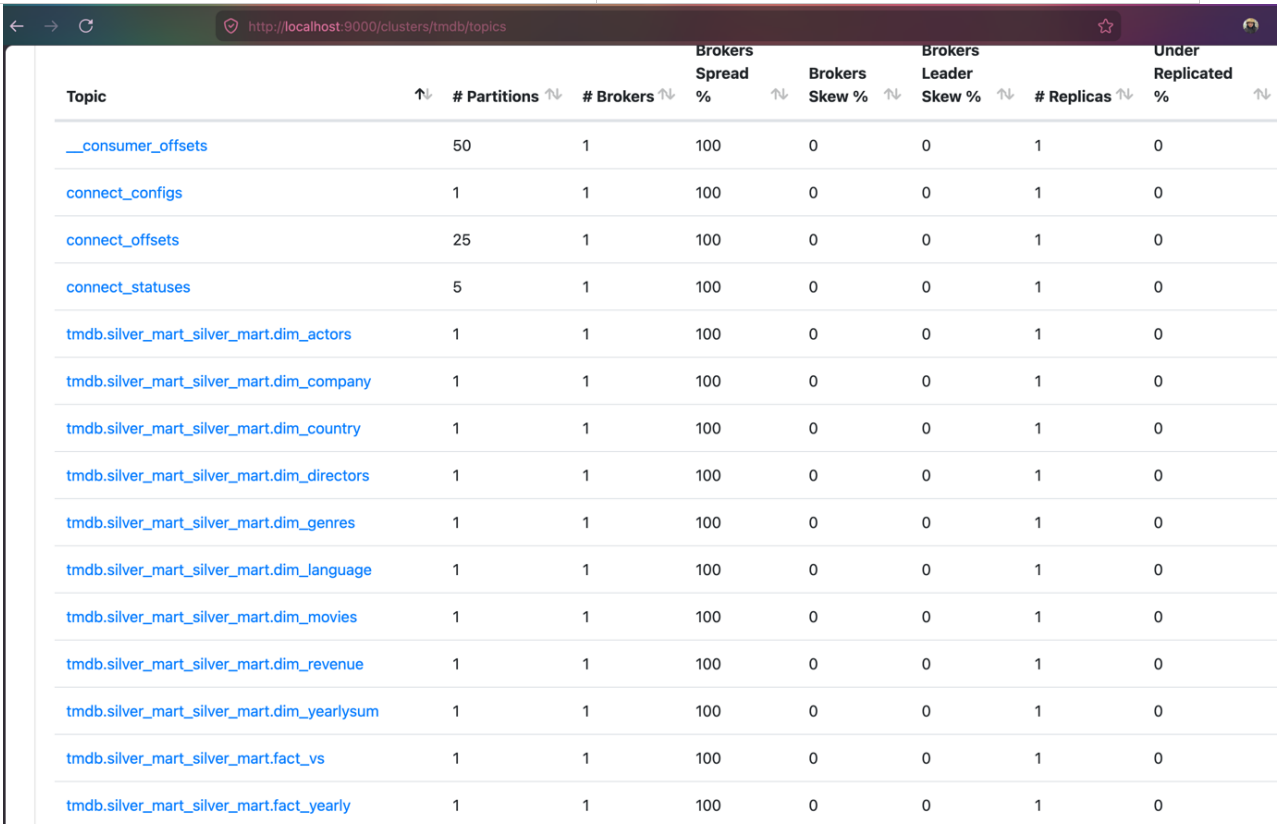
```
# Masuk ke container PostgreSQL
docker exec -it tmdb_postgres bash

# Set WAL level ke logical
echo "wal_level = logical" >> /var/lib/postgresql/data/postgresql.conf

# Restart container
docker restart tmdb_postgres

-- Verifikasi di pgAdmin:
SHOW wal_level;  -- harus menampilkan: logical

-- Cek publikasi Debezium:
SELECT * FROM pg_publication;
-- pubname: dbz_publication | puballtables: true
-- pubinsert/pubupdate/pubdelete: true
```

6.3 Hasil: 16 Kafka Topics

Debezium secara otomatis membuat 1 topic per tabel yang di-capture:

Topic Kafka	Sumber Tabel

tmdb.silver_mart.dim_actor	silver_mart.dim_actor
tmdb.silver_mart.dim_company	silver_mart.dim_company
tmdb.silver_mart.dim_country	silver_mart.dim_country
tmdb.silver_mart.dim_directors	silver_mart.dim_directors
tmdb.silver_mart.dim_genres	silver_mart.dim_genres
tmdb.silver_mart.dim_language	silver_mart.dim_language
tmdb.silver_mart.dim_movies	silver_mart.dim_movies
tmdb.silver_mart.dim_revenue	silver_mart.dim_revenue
tmdb.silver_mart.dim_yearlysum	silver_mart.dim_yearlysum
tmdb.silver_mart.fact_vs	silver_mart.fact_vs
tmdb.silver_mart.fact_yearly	silver_mart.fact_yearly
+ 5 internal Kafka topics	__consumer_offsets, connect_configs, dll.



Topic	# Partitions	# Brokers	Brokers Spread %	Brokers Skew %	Brokers Leader Skew %	# Replicas	Under Replicated %
__consumer_offsets	50	1	100	0	0	1	0
connect_configs	1	1	100	0	0	1	0
connect_offsets	25	1	100	0	0	1	0
connect_statuses	5	1	100	0	0	1	0
tmdb.silver_mart_silver_mart.dim_actors	1	1	100	0	0	1	0
tmdb.silver_mart_silver_mart.dim_company	1	1	100	0	0	1	0
tmdb.silver_mart_silver_mart.dim_country	1	1	100	0	0	1	0
tmdb.silver_mart_silver_mart.dim_directors	1	1	100	0	0	1	0
tmdb.silver_mart_silver_mart.dim_genres	1	1	100	0	0	1	0
tmdb.silver_mart_silver_mart.dim_language	1	1	100	0	0	1	0
tmdb.silver_mart_silver_mart.dim_movies	1	1	100	0	0	1	0
tmdb.silver_mart_silver_mart.dim_revenue	1	1	100	0	0	1	0
tmdb.silver_mart_silver_mart.dim_yearlysum	1	1	100	0	0	1	0
tmdb.silver_mart_silver_mart.fact_vs	1	1	100	0	0	1	0
tmdb.silver_mart_silver_mart.fact_yearly	1	1	100	0	0	1	0

Step 7 — ClickHouse OLAP: Column-Oriented Analytics

ClickHouse adalah database OLAP column-oriented yang jauh lebih cepat untuk analitik agregasi dibanding PostgreSQL (row-oriented). Di sinilah Medallion Architecture diimplementasikan ulang dalam ClickHouse: Bronze (Kafka Engine) → Silver (ReplacingMergeTree) → Gold (View).

Cek Clickhouse di : http://localhost:18123/play

7.1 Perbedaan PostgreSQL vs ClickHouse

Aspek	PostgreSQL (OLTP)	ClickHouse (OLAP)
Cara simpan	Per baris (row-oriented)	Per kolom (column-oriented)
Kecepatan analitik	Lebih lambat untuk agregasi	Jauh lebih cepat untuk analisis kolom
Use case	Transaksi: order, banking	Analitik: BI, Spotify Wrapped
MV auto-update	Tidak — perlu trigger manual	Ya — otomatis tanpa trigger

7.2 Arsitektur 3 Layer di ClickHouse

BRONZE: Kafka Engine Table (dim_movies_queue) — Langsung consume dari Kafka topic via ENGINE = Kafka.

SILVER: ReplacingMergeTree Table (dim_movies) — Simpan data final, hindari duplikat via ENGINE = ReplacingMergeTree(insert_time).

GOLD: View (tmdb_movies_gold) — Tambah kolom kalkulasi: profit, performance_label, popularity_tier. WHERE budget > 0 AND revenue > 0.

7.3 Objek di ClickHouse (9 Objek Total)

Nama Objek	Tipe	Engine / Keterangan
dim_movies	Base Table	ReplacingMergeTree — data final silver layer
dim_movies_queue	Kafka Queue	Kafka Engine — consume dari topic
dim_movies_mv	Materialized View	Jembatan queue → base table (otomatis)
dim_revenue	Base Table	ReplacingMergeTree
dim_revenue_queue	Kafka Queue	Kafka Engine
dim_revenue_mv	Materialized View	Otomatis
fact_yearly	Base Table	ReplacingMergeTree
fact_yearly_queue	Kafka Queue	Kafka Engine

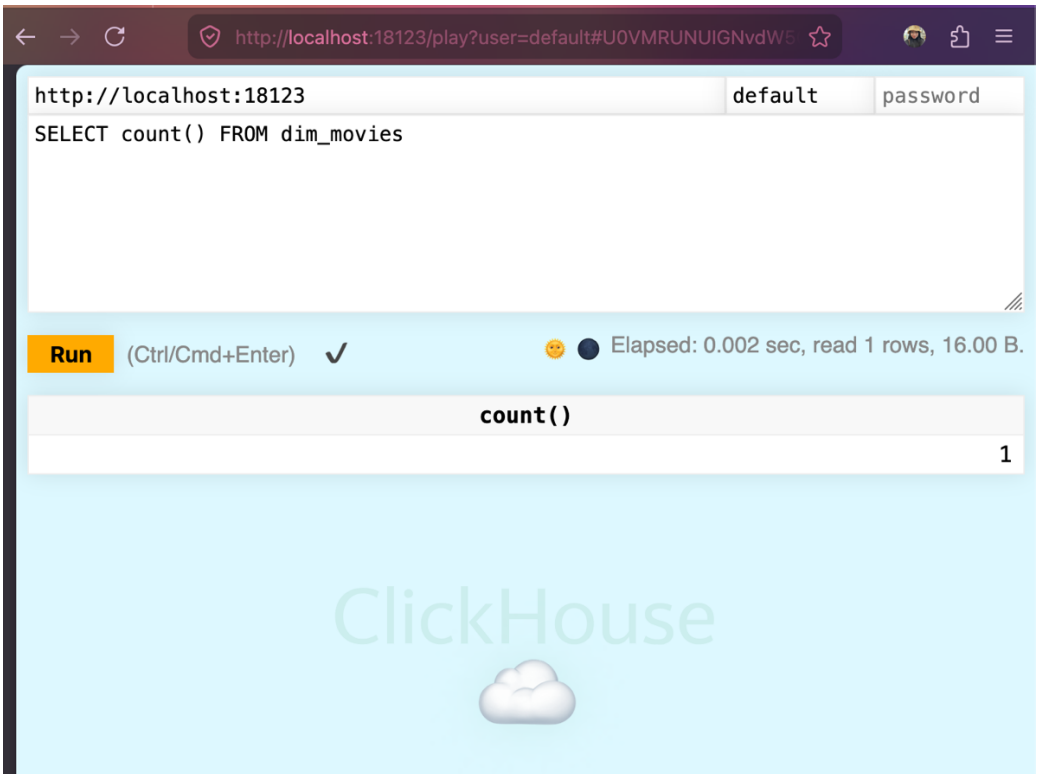
fact_yearly_mv	Materialized View	Otomatis

```
macbookpro@MacBook-Pro-MacBook tmdb-pipeline-project % curl http://localhost:18123/ --data-binary "SHOW TABLES"
dim_movies
dim_movies_mv
dim_movies_queue
dim_revenue
dim_revenue_mv
dim_revenue_queue
fact_yearly
fact_yearly_mv
fact_yearly_queue
macbookpro@MacBook-Pro-MacBook tmdb-pipeline-project %
```

Ln 54, Col 34 (1512 selected) Spaces: 4 UTF-8 LF { } MS SQL

7.4 Hasil Validasi Data di ClickHouse

Tabel	Jumlah Rows	Keterangan
dim_movies	4.802 rows	Data mengalir dari Kafka CDC
dim_revenue	3.229 rows	Film dengan data keuangan lengkap
fact_yearly	363 rows	Agregasi per tahun per genre



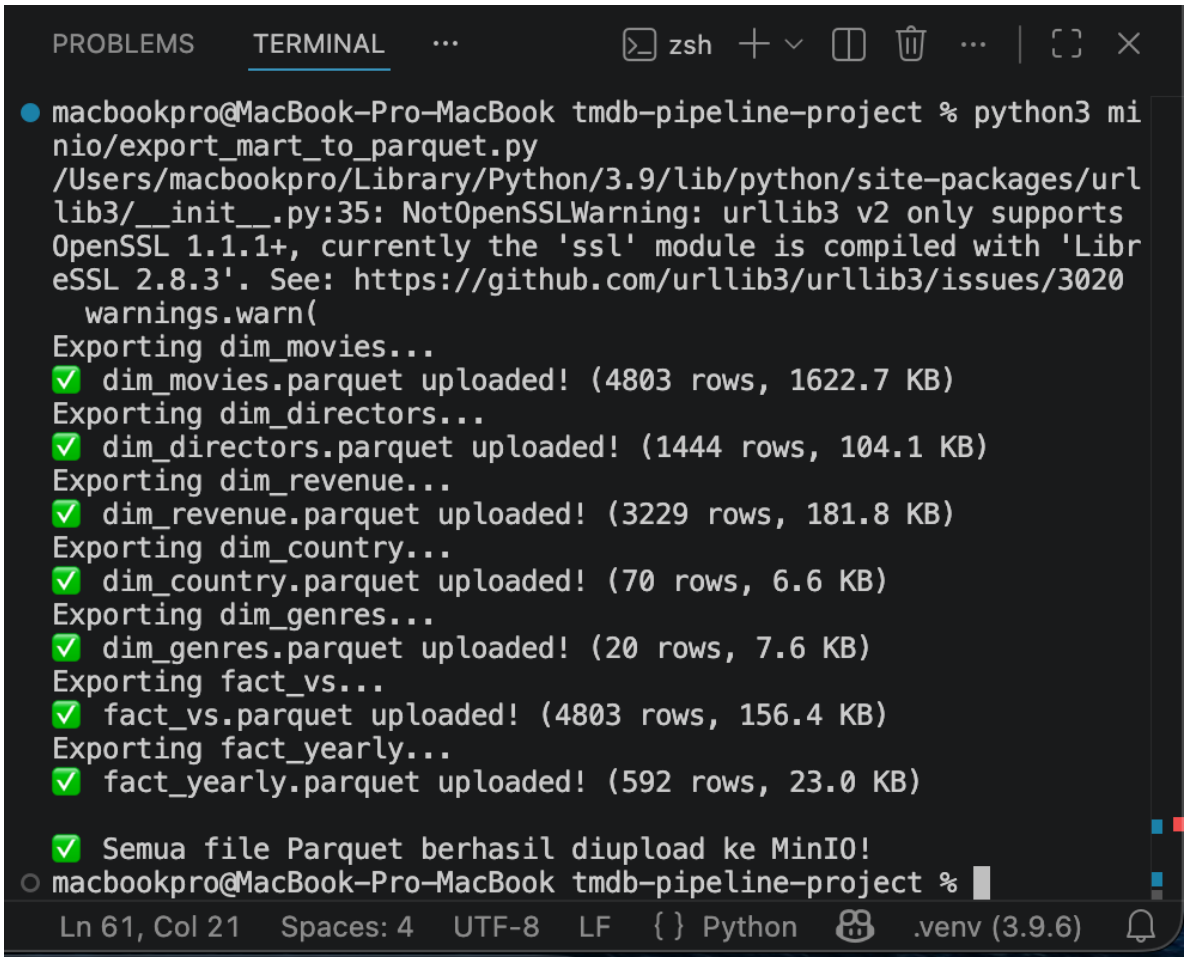
Step 8 — MinIO Object Storage: Modern HDFS Replacement

MinIO adalah object storage modern yang kompatibel dengan Amazon S3, digunakan untuk menyimpan hasil mart layer dalam format Parquet.

Masuk ke MINIO = http://localhost:9001/browser loginnnya = minioadmin/minioadmin

8.1 Konfigurasi

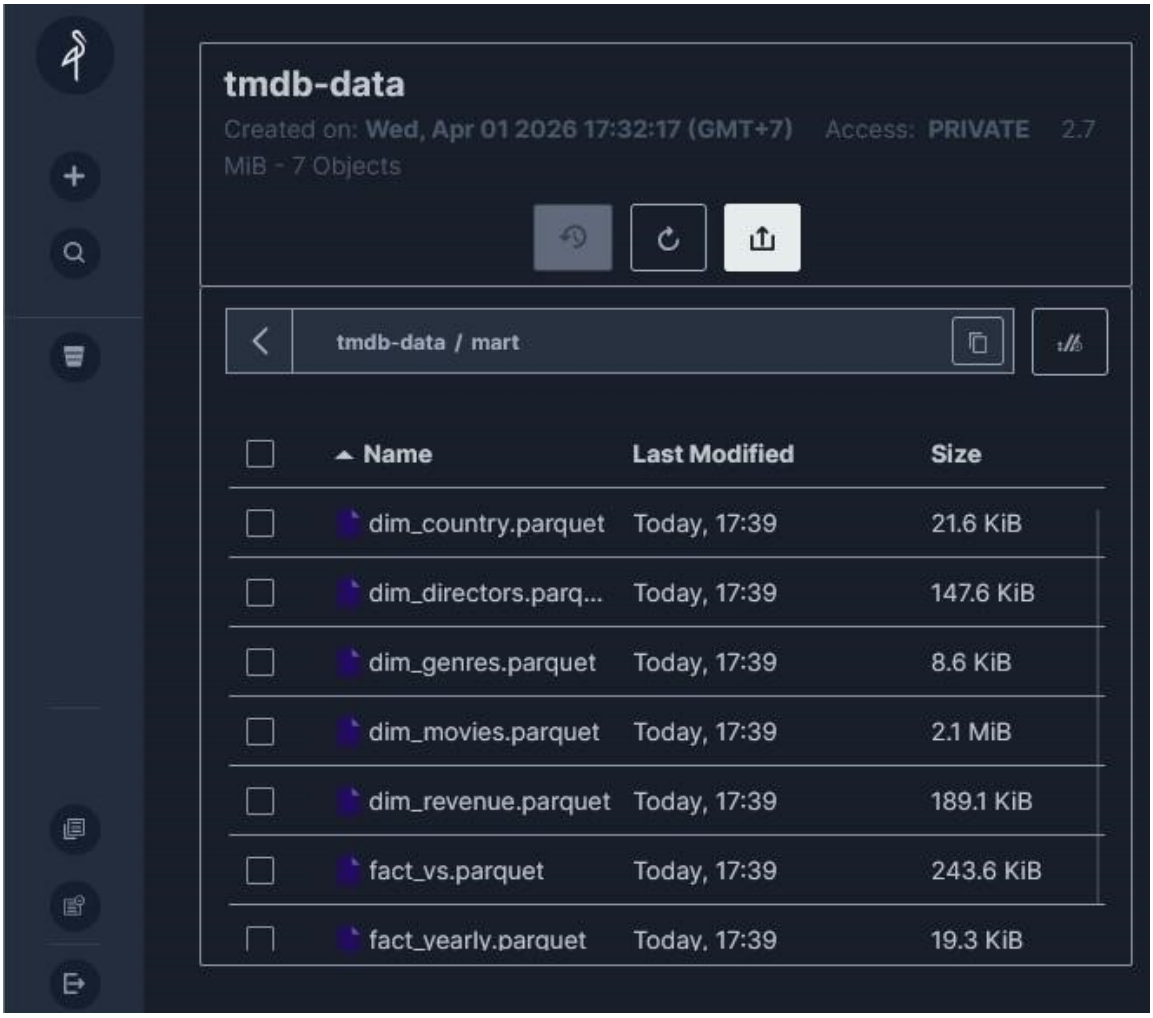
```
docker run -d -p 9002:9000 -p 9001:9001 --name tmdb_minio \
  -v C:\data\tmdb-minio:/data \
  -e MINIO_ROOT_USER=minioadmin \
  -e MINIO_ROOT_PASSWORD=minioadmin \
  quay.io/minio/minio server /data --console-address ':9001'


Akses Web UI : http://localhost:9001
Username     : minioadmin
Password     : minioadmin
```



8.2 Hasil Upload — Bucket: tmdb-data/mart/

Total: 7 file Parquet, 2.7 MiB

File Parquet	Jumlah Rows	Ukuran	Tanggal Upload
dim_movies.parquet	4.802	2.1 MiB	01 Apr 2026, 17:39
dim_directors.parquet	2.349	147.6 KiB	01 Apr 2026, 17:39
fact_vs.parquet	4.867	243.6 KiB	01 Apr 2026, 17:39
dim_revenue.parquet	3.229	189.1 KiB	01 Apr 2026, 17:39
dim_country.parquet	70	21.6 KiB	01 Apr 2026, 17:39
fact_yearly.parquet	363	19.3 KiB	01 Apr 2026, 17:39
dim_genres.parquet	19	8.6 KiB	01 Apr 2026, 17:39



Step 9 — Streamlit Dashboard: Visualisasi Data

Dashboard interaktif dengan Streamlit (Python library) yang terhubung langsung ke data di PostgreSQL. Dashboard menampilkan scorecard KPI, chart, dan tabel dengan filter interaktif.

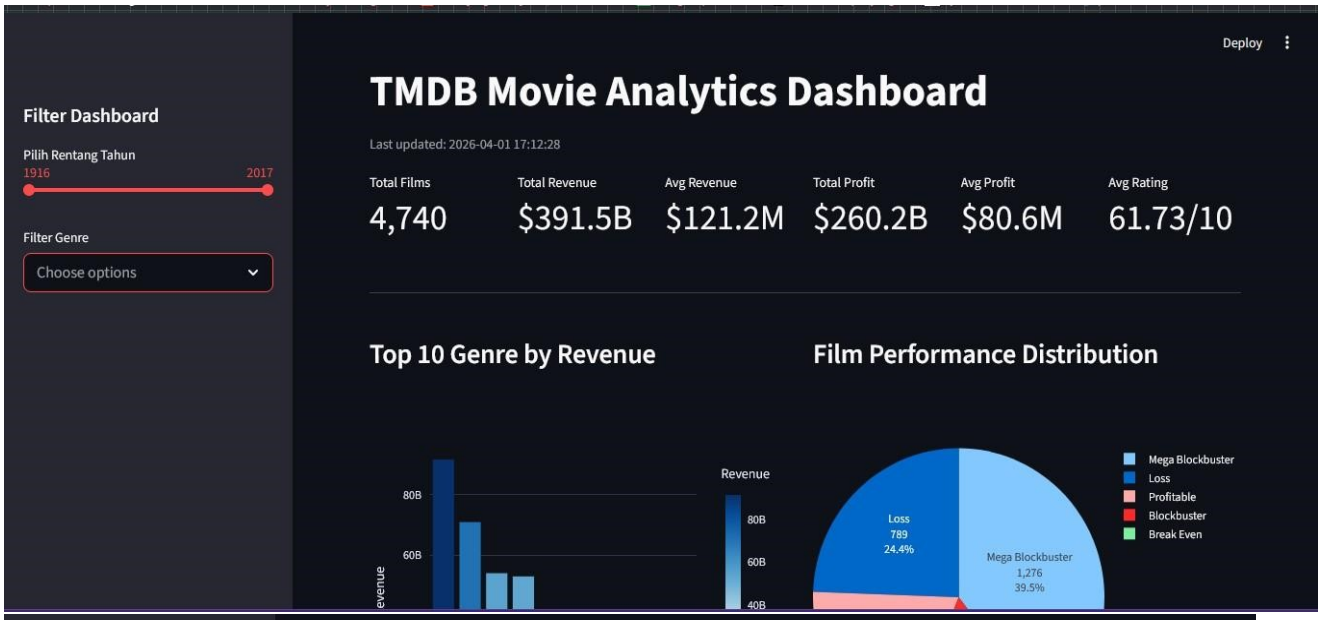
Cek dashboard Streamlit = http://localhost:8501/

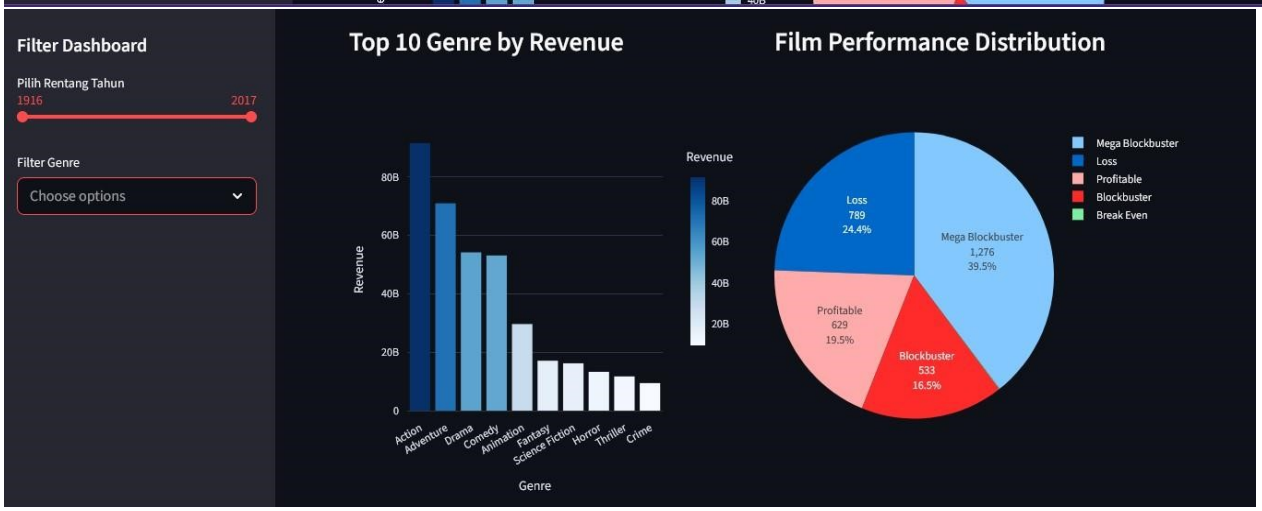
9.1 Fitur Dashboard

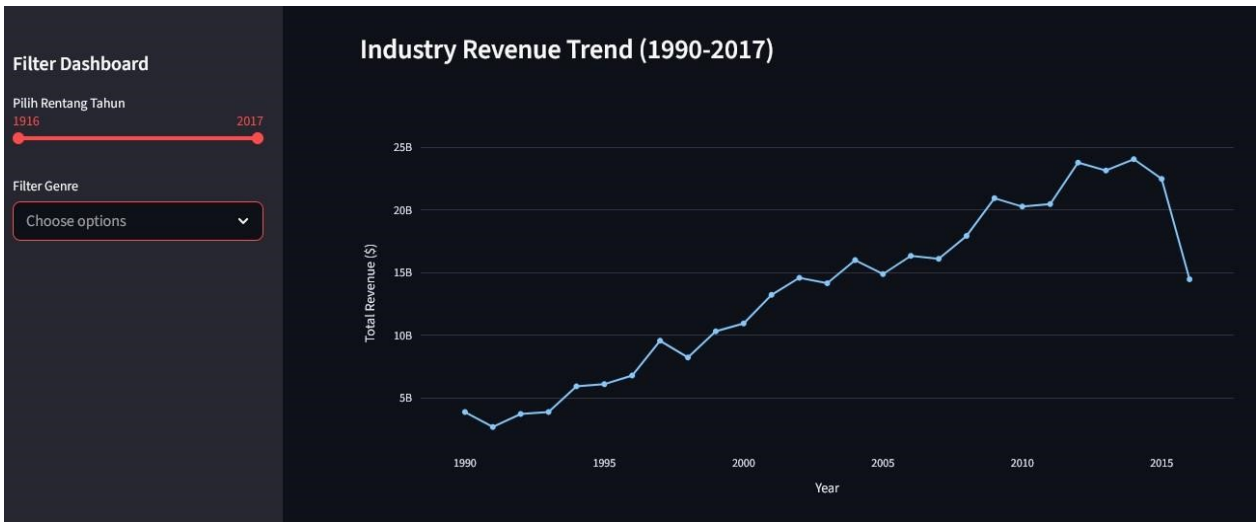
Fitur	Keterangan
Filter Rentang Tahun	Slider untuk memilih tahun rilis (misal: 1959–2004)
Filter Genre	Multi-select untuk memfilter genre (Comedy, Adventure, dll.)
Scorecard KPI	Total Films, Total Revenue, Avg Revenue, Total Profit, Avg Profit, Avg Rating
Top 10 Genre by Revenue	Bar chart revenue per genre
Film Performance Distribution	Pie chart distribusi Mega Blockbuster / Blockbuster / Profitable / Loss
Top 20 Highest Revenue Films	Tabel detail dengan kolom Title, Genre, Lead Actor, Director, Revenue, Performance

9.2 Contoh Data Dashboard

> *Filter: 1959–2004, Genre: Comedy + Adventure — Film tertinggi dalam filter ini: The Lord of the Rings: The Return of the King ($1.118B revenue, Adventure, sutradara Peter Jackson)*


Deploy
TMDB Movie Analytics Dashboard
Filter Dashboard
Pilih Rentang Tahun
1916
2017
Filter Genre
Choose options
Last updated: 2026-04-01 17:12:28
Total Films
4,740
Total Revenue
$391.5B
Avg Revenue
$121.2M
Total Profit
$260.2B
Avg Profit
$80.6M
Avg Rating
61.73/10
Top 10 Genre by Revenue
Film Performance Distribution
Revenue
80B
60B
40B
Loss
789
24.4%
Mega Blockbuster
1,276
39.5%
Mega Blockbuster
Loss
Profitable
Blockbuster
Break Even

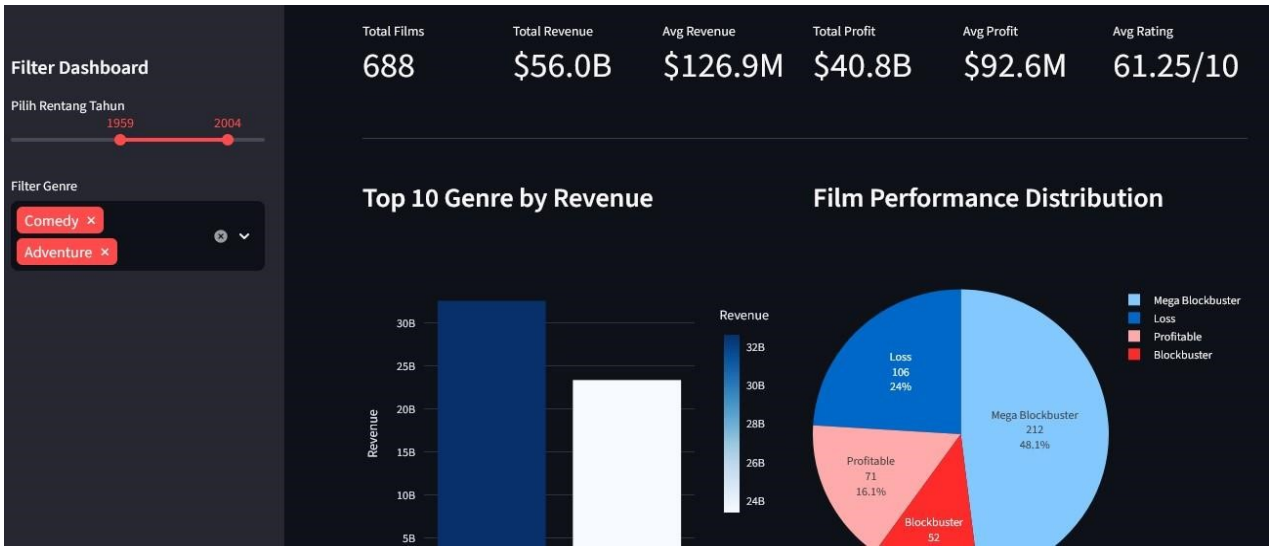

Filter Dashboard
Pilih Rentang Tahun
1916
2017
Filter Genre
Choose options
Top 10 Genre by Revenue
Film Performance Distribution
Revenue
80B
60B
40B
20B
0
Action
Adventure
Drama
Comedy
Animation
Fantasy
Science Fiction
Horror
Thriller
Crime
Genre
Loss
789
24.4%
Mega Blockbuster
1,276
39.5%
Profitable
629
19.5%
Blockbuster
533
16.5%
Mega Blockbuster
Loss
Profitable
Blockbuster
Break Even


Filter Dashboard
Pilih Rentang Tahun
1916
2017
Filter Genre
Choose options
Industry Revenue Trend (1990-2017)
Total Revenue ($)
25B
20B
15B
10B
5B
1990
1995
2000
2005
2010
2015
Year

	Title	Genre	Lead Actor	Director	Revenue	Profit	Performance	Ratin
0	Avatar	Action	Sam Worthington	James Cameron	$2,787,965,087	$2,550,965,087	Mega Blockbuster	7.2/1
1	Titanic	Drama	Kate Winslet	James Cameron	$1,845,034,188	$1,645,034,188	Mega Blockbuster	7.5/1
2	The Avengers	Science Fiction	Robert Downey Jr.	Joss Whedon	$1,519,557,910	$1,299,557,910	Mega Blockbuster	7.4/1
3	Jurassic World	Action	Chris Pratt	Colin Trevorrow	$1,513,528,810	$1,363,528,810	Mega Blockbuster	6.5/1
4	Furious 7	Action	Vin Diesel	James Wan	$1,506,249,360	$1,316,249,360	Mega Blockbuster	7.3/1
5	Avengers: Age of Ultron	Action	Robert Downey Jr.	Joss Whedon	$1,405,403,694	$1,125,403,694	Mega Blockbuster	7.3/1
6	Frozen	Animation	Kristen Bell	Chris Buck	$1,274,219,009	$1,124,219,009	Mega Blockbuster	7.3/1
7	Iron Man 3	Action	Robert Downey Jr.	Shane Black	$1,215,439,994	$1,015,439,994	Mega Blockbuster	6.8/1
8	Minions	Family	Sandra Bullock	Kyle Balda	$1,156,730,962	$1,082,730,962	Mega Blockbuster	6.4/1
9	Captain America: Civil W	Adventure	Chris Evans	Anthony Russo	$1,153,304,495	$903,304,495	Mega Blockbuster	7.1/1

9.2 Contoh Data Dashboard (Filter: 1959–2004, Genre: Comedy + Adventure)

Film tertinggi dalam filter ini: The Lord of the Rings: The Return of the King ($1.118B revenue, Adventure, Peter Jackson)



	Title	Genre	Lead Actor	Director	Revenu		
0	The Lord of the Rings: The Re	Adventure	Elijah Wood	Peter Jackson	$1,118,		
1	Harry Potter and the Philosop	Adventure	Daniel Radcliffe	Chris Columbus	$976,475,550	$851,475,550	Mega Blockbus 7.5/10
2	The Lord of the Rings: The Tw	Adventure	Elijah Wood	Peter Jackson	$926,287,400	$847,287,400	Mega Blockbus 8.0/10
3	Star Wars: Episode I - The Pha	Adventure	Liam Neeson	George Lucas	$924,317,558	$809,317,558	Mega Blockbus 6.3/10
4	Jurassic Park	Adventure	Sam Neill	Steven Spielberg	$920,100,000	$857,100,000	Mega Blockbus 7.6/10
5	Shrek 2	Adventure	Mike Myers	Andrew Adamson	$919,838,758	$769,838,758	Mega Blockbus 6.7/10
6	Harry Potter and the Chambe	Adventure	Daniel Radcliffe	Chris Columbus	$876,688,482	$776,688,482	Mega Blockbus 7.4/10
7	The Lord of the Rings: The Fel	Adventure	Elijah Wood	Peter Jackson	$871,368,364	$778,368,364	Mega Blockbus 8.0/10
8	Harry Potter and the Prisoner	Adventure	Daniel Radcliffe	Alfonso Cuarón	$789,804,554	$659,804,554	Mega Blockbus 7.7/10
9	Star Wars	Adventure	Mark Hamill	George Lucas	$775,398,007	$764,398,007	Mega Blockbus 8.1/10

Step 10 — Airflow DAG Orchestration

Apache Airflow digunakan untuk mengotomasi dan menjadwalkan jalannya pipeline TMDB secara end-to-end. DAG (Directed Acyclic Graph) bernama tmdb_pipeline_dag mengatur urutan eksekusi 5 task secara berurutan dan terjadwal otomatis.

Akses Airflow UI: http://localhost:8082

9.1 Konfigurasi DAG

Komponen	Nilai
DAG ID	tmdb_pipeline_dag
Deskripsi	TMDB End-to-End Pipeline
Schedule	0 0 * * 1 (setiap Senin tengah malam)
Next Run	2026-06-01, 00:00:00
Catchup	False
Owner	nabila
Retries per Task	1 (retry delay: 5 menit)

9.2 Urutan Task dalam DAG

Urutan	Task ID	Fungsi
1	fetch_tmdb_api	Ambil data film terbaru dari TMDb API live → simpan ke tmdb_raw
2	load_kaggle_csv	Load file CSV Kaggle ke PostgreSQL (csv_movies_raw, csv_credits_raw)
3	merge_raw	Gabungkan kedua sumber menjadi tabel merge_raw (4.803 rows)
4	export_to_minio	Export 7 tabel mart dari PostgreSQL → MinIO format Parquet
5	export_to_bigquery	Baca 7 Parquet dari MinIO → Upload ke Google BigQuery (tmdb_gold)

Alur eksekusi: fetch_tmdb_api → load_kaggle_csv → merge_raw → export_to_minio → export_to_bigquery

Eksekusi bersifat sequential — jika satu task gagal, task berikutnya tidak dijalankan.

9.3 Hasil Eksekusi DAG

Metrik	Nilai
Status Run Terakhir	✅ Success (semua 5 task hijau)
Timestamp Run Sukses	2026-06-02, 03:40:04 UTC
Durasi Maksimum	00:05:53

Hasil per Task (Run Terakhir):

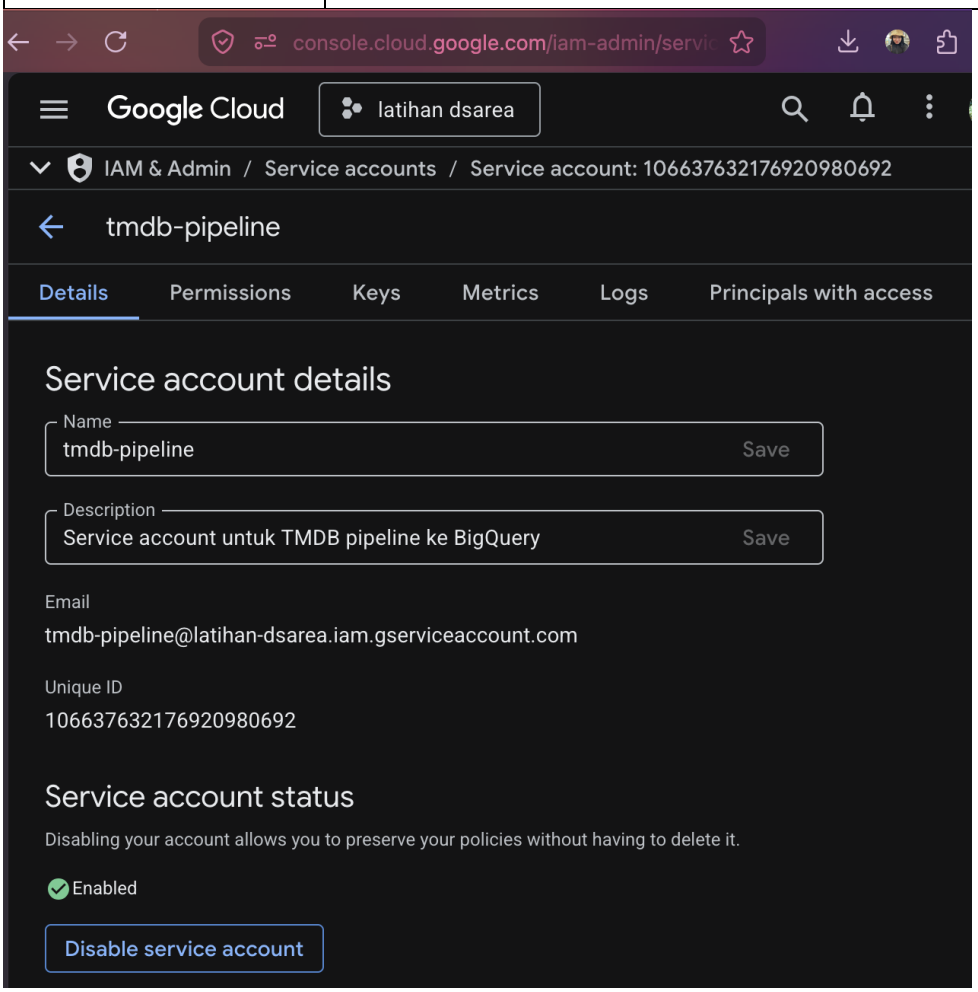
Task	Status	Output
fetch_tmdb_api	✅ Success	80 rows → tmdb_raw
load_kaggle_csv	✅ Success	4.673 + 4.801 rows → PostgreSQL

merge_raw	✅ Success	4.803 rows → merge_raw
export_to_minio	✅ Success	7 file Parquet → tmdb-data/mart/
export_to_bigquery	✅ Success	7 tabel → BigQuery dataset tmdb_gold

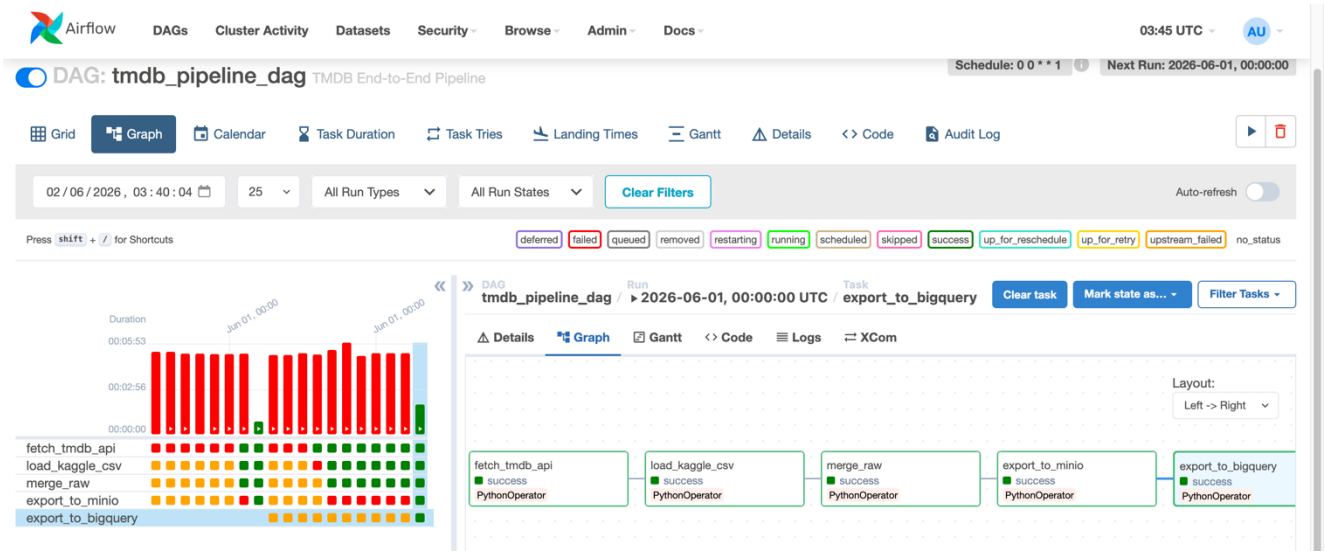
Catatan Development: Kegagalan run sebelumnya terjadi selama fase konfigurasi environment — penyebab utama meliputi koneksi PostgreSQL menggunakan localhost (seharusnya nama container tmdb_postgres), MinIO endpoint menggunakan nama container dengan underscore yang tidak kompatibel dengan boto3 (diselesaikan dengan IP address 172.18.0.2:9000 + Config(signature_version='s3v4')), dan schema dbt yang perlu disesuaikan dari silver_mart menjadi silver_mart_silver_mart. Run terakhir berhasil penuh setelah semua konfigurasi diperbaiki.

9.4 Yang Dikontrol Airflow vs Berjalan Mandiri

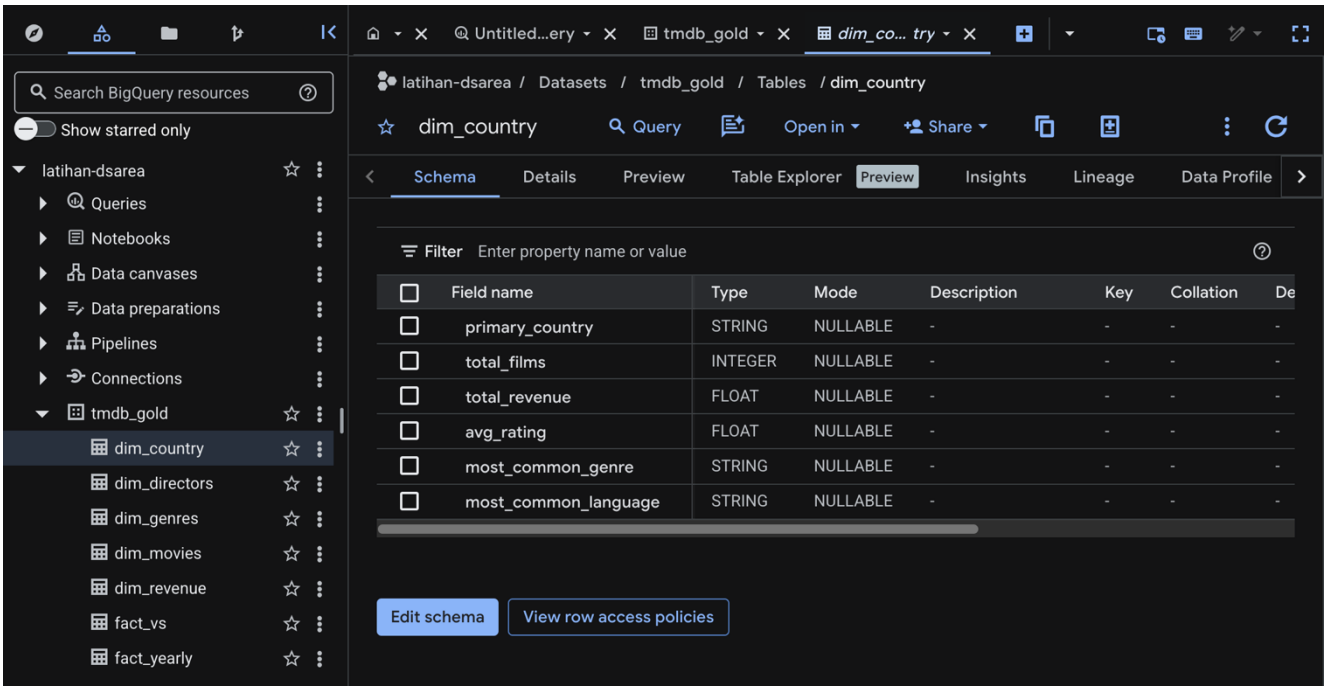
Komponen	Keterangan
✅ Dikontrol DAG	fetch_tmdb_api → load_kaggle_csv → merge_raw → export_to_minio → export_to_bigquery
⚡ Berjalan Mandiri	Kafka + Debezium (real-time CDC), ClickHouse (auto-consume), Streamlit (always on)



Hasil di Airflow = 5 task berhasil



Hasil di Bigquery, 7 file dari Minio berhasil di ekspor.



Lampiran: Ringkasan Semua Objek Database

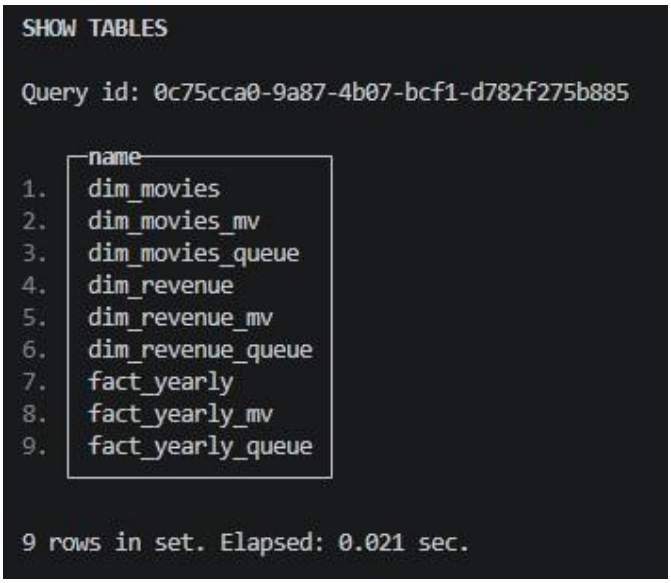
1) PostgreSQL — tmdb_db

Schema	Tabel / Objek	Tipe	Keterangan
public	merge_raw	Tabel	Bronze: 4.803 rows, 22 kolom, hasil merge 2 CSV
public	tmdb_raw	Tabel	Bronze: 80 rows dari TMDb API
public	fact_yearly_partisi	Tabel Induk	Partisi RANGE berdasarkan release_year
public	fact_yearly_1920s … 2010s	Partisi (10x)	Sub-tabel per dekade
public	pyspark_tmdb_finance	Tabel	Hasil PySpark ETL — per film dengan kalkulasi
public	pyspark_genre_aggregation	Tabel	Hasil PySpark ETL — agregasi per genre (20 rows)
public	pyspark_tmdb_result	Tabel	Hasil PySpark ETL awal (versi sederhana)
public	nifi_tmdb_movie	Tabel	Hasil Apache NiFi pipeline dari CSV
silver_mart	dim_movies	Tabel dbt	Master film lengkap — 4.802 rows
silver_mart	dim_genres	Tabel dbt	Ringkasan per genre — 19 rows
silver_mart	dim_directors	Tabel dbt	Profil sutradara — 2.349 rows
silver_mart	dim_actor	Tabel dbt	Profil aktor
silver_mart	dim_country	Tabel dbt	Ringkasan per negara — 70 rows
silver_mart	dim_language	Tabel dbt	Ringkasan per bahasa
silver_mart	dim_company	Tabel dbt	Profil rumah produksi
silver_mart	dim_revenue	Tabel dbt	Fakta keuangan — 3.229 rows
silver_mart	dim_yearlysum	Tabel dbt	Ringkasan tahunan
silver_mart	fact_vs	Tabel dbt	Rekonsiliasi 2 sumber — 4.867 rows
silver_mart	fact_yearly	Tabel dbt	Tren per tahun per genre — 363 rows
public	mv_genre_summary	MV	Cache agregasi genre
public	mv_revenue_summary	MV	Cache analisis keuangan per dekade
public	mv_director_performance	MV	Cache profil sutradara
public	mv_country_performance	MV	Cache profil negara

2) ClickHouse — 9 Objek

Nama	Tipe	Engine	Rows
dim_movies	Base Table	ReplacingMergeTree	9.604
dim_movies_queue	Kafka Table	Kafka Engine	—
dim_movies_mv	Materialized View	—	—

dim_revenue	Base Table	ReplacingMergeTree	6.458
dim_revenue_queue	Kafka Table	Kafka Engine	—
dim_revenue_mv	Materialized View	—	—
fact_yearly	Base Table	ReplacingMergeTree	726
fact_yearly_queue	Kafka Table	Kafka Engine	—
fact_yearly_mv	Materialized View	—	—



3) Container Docker yang Berjalan

Service	Container Name	Image	Port
Airflow Scheduler	tmdb_airflow_scheduler	apache/airflow:2.8.1	-
Airflow Webserver	tmdb_airflow_web	apache/airflow:2.8.1	8082:8080
Airflow Init	tmdb_airflow_init	apache/airflow:2.8.1	-
Debezium Connect	tmdb_debezium	debezium/connect:2.4	8083:8083
pgAdmin	tmdb_pgadmin	dpage/pgadmin4	5050:80
Kafka Broker	tmdb_kafka	confluentinc/cp-kafka:7.4.0	29092:29092
Kafka Manager	tmdb_kafka_man	hlebalbau/kafka-manager:latest	9000:9000
PostgreSQL	tmdb_postgres	postgres:13	5432:5432
ClickHouse	tmdb_clickhouse	clickhouse/clickhouse-server:23.8	18123:8123
ZooKeeper	tmdb_zookeeper	confluentinc/cp-zookeeper:7.4.0	-
MinIO	tmdb_minio	minio/minio:latest	9002:9000